

Reduced-Form CDS Pricing: Bootstrapping & Mark-to-Market

1 Methodology and Code

1.1 Term Structure of Interest Rates and Discount Factors

We use the Nelson-Siegel model to construct the continuously compounded interest rate curve. Given parameters $\beta_0 = 0.0408$, $\beta_1 = -0.0396$, $\beta_2 = -0.0511$, and $\tau = 1.614$, the interest rate at maturity s is:

$$\hat{r}(s) = \beta_0 + \beta_1 \frac{\tau}{s} \left(1 - e^{-s/\tau}\right) + \beta_2 \left[\frac{\tau}{s} \left(1 - e^{-s/\tau}\right) - e^{-s/\tau}\right] \quad (1)$$

The discount factor is then:

$$Z(0, s) = e^{-\hat{r}(s) \cdot s} \quad (2)$$

These are implemented in the functions `nelson_siegel_rate()` and `discount_factor()`:

```
def nelson_siegel_rate(s, beta0, beta1, beta2, tau):
    term1 = beta0
    term2 = beta1 * (tau / s) * (1 - np.exp(-s / tau))
    term3 = beta2 * ((tau/s)*(1 - np.exp(-s/tau)) - np.exp(-s/tau))
    return term1 + term2 + term3

def discount_factor(s, beta0, beta1, beta2, tau):
    r = nelson_siegel_rate(s, beta0, beta1, beta2, tau)
    return np.exp(-r * s)
```

Note that equation (1) contains the term τ/s , which is singular at $s = 0$. Since the protection leg grid and survival probability computations require evaluation at $s = 0$ (where $Z(0, 0) = 1$ and $Q(0, 0) = 1$ by definition), the notebook provides a variant `discount_factor_safe()` that handles this boundary case:

```
def discount_factor_safe(t, beta0, beta1, beta2, tau):
    if np.isscalar(t) and t == 0:
        return 1.0 # Z(0,0) = exp(0) = 1
    r = nelson_siegel_rate_safe(t, beta0, beta1, beta2, tau)
    return np.exp(-r * np.asarray(t))
```

Similarly, `nelson_siegel_rate_safe()` returns $\beta_0 + \beta_1$ at $s = 0$, which is the correct limiting value of $\hat{r}(s)$ as $s \rightarrow 0^+$. The bootstrap (Section 1.3) and MTM calculations (Section 1.4) use these safe variants throughout.

1.2 Survival Probability Curve

The survival probability is modelled as a piecewise-exponential function driven by a piecewise-constant forward default rate $h(t) = h_k$ for $t \in (T_{k-1}, T_k]$, where $T_0 = 0$. For $t \in (T_{k-1}, T_k]$:

$$Q(0, t) = \exp\left(-\sum_{i=1}^{k-1} (T_i - T_{i-1}) h_i - (t - T_{k-1}) h_k\right) \quad (3)$$

This is implemented in `survival_probability_pieewise()`:

```

def survival_probability_pieewise(t, maturities, h_values):
    total = 0.0
    T_prev = 0.0
    for T_i, h_i in zip(maturities, h_values):
        if t <= T_i:
            total += (t - T_prev) * h_i
            return np.exp(-total)
        total += (T_i - T_prev) * h_i
        T_prev = T_i
    return np.exp(-total)

```

The hazard rates $h_1, \dots, h_M$ are calibrated via bootstrapping (Section 1.3).

1.3 Calibration via Newton-Raphson Bootstrap

We calibrate the hazard rates $h_1, \dots, h_M$ sequentially from the 10 market CDS spreads in Table 1 of the assignment. For each maturity T_i , we solve $V_i(h_i) = 0$ where:

$$V_i(h_i) = \frac{1 - R_i}{2} \sum_{k=1}^{K_i} (Z(0, s_{k-1}) + Z(0, s_k)) (Q(0, s_{k-1}) - Q(0, s_k)) - S_i \times \text{RPV01}_i \quad (4)$$

with $s_k = kT_i/K_i$, $K_i = \lceil mT_i \rceil$, and $m = 12$ monthly subintervals per year. The RPV01 for an on-market CDS is:

$$\text{RPV01}_i = \frac{1}{2} \sum_{n=1}^{N_i} \Delta(t_{n-1}^i, t_n^i) Z(0, t_n^i) (Q(0, t_{n-1}^i) + Q(0, t_n^i)) \quad (5)$$

The Newton-Raphson iteration is $x_{k+1} = x_k - V_i(x_k)/V_i'(x_k)$, starting from $x_0 = 0.01$ with tolerance $\delta = 10^{-6}$. We compute the analytic derivative using:

$$\frac{\partial Q(0, t)}{\partial h_i} = \begin{cases} 0, & t \leq T_{i-1} \\ -(t - T_{i-1}) Q(0, t), & T_{i-1} < t \leq T_i \end{cases} \quad (6)$$

The core Newton-Raphson loop is implemented in `solve_hi_newton()`:

```

def solve_hi_newton(i, known_h, maturities, spreads,
                    recovery_rate, m, x0=0.01, tol=1e-6):
    x = x0
    for iteration in range(max_iter):
        fx = cds_equation(x, i, known_h, ...)
        dfx = cds_equation_derivative(x, i, known_h, ...)
        if abs(fx) < tol:
            return x # converged
        x = x - fx / dfx # Newton update
    return x

```

The analytic derivative is computed in `dQ_dhi()`:

```

def dQ_dhi(t, i, maturities, h_values):
    T_prev = 0.0 if i == 0 else maturities[i - 1]
    if t <= T_prev:
        return 0.0
    q = survival_probability_pieewise(t, maturities, h_values)
    return -(t - T_prev) * q

```

The bootstrap proceeds from shortest to longest maturity, with previously calibrated h values held fixed at each step.

1.4 Mark-to-Market Value of Non-Standard CDS

The non-standard CDS is a seasoned contract—it was entered into on 15 May 2012 and is being valued as of 28 February 2014, approximately two years into its life. The contract details are summarised in Table 1.

Table 1: Non-standard CDS contract parameters

Parameter	Value
Face Value	USD 100 million (long protection)
Valuation Date	28 February 2014
Effective Date	15 May 2012
Maturity Date	15 May 2042
Contractual Spread	170 bp per annum
Expected Recovery Rate	60%
Payment Frequency	Quarterly (15 Feb, 15 May, 15 Aug, 15 Nov)
Day Count Convention	Actual/360
Business Day Convention	Modified Following

Key differences from the standard CDS used in calibration: the recovery rate is 60% (versus 45%), the payment dates fall on the 15th of Feb/May/Aug/Nov (versus the 20th of Mar/Jun/Sep/Dec), and the maturity date (15 May 2042) does not fall on a standard CDS date (20 Mar/Jun/Sep/Dec).

The mark-to-market value for the protection buyer is:

$$V(0) = \frac{1-R}{2} \sum_{k=1}^K (Z(0, s_{k-1}) + Z(0, s_k)) (Q(0, s_{k-1}) - Q(0, s_k)) - S_{-t} \times \text{RPV01}(0, T) \quad (7)$$

The protection leg is implemented in `protection_leg()`:

```
def protection_leg(ns_maturity_years, ns_recovery,
                  all_maturities, h_values, m=12):
    K = int(np.ceil(m * ns_maturity_years))
    epsilon = ns_maturity_years / K
    prot = 0.0
    for k in range(1, K + 1):
        s_prev, s_curr = (k-1)*epsilon, k*epsilon
        Z_prev = discount_factor_safe(s_prev, ...)
        Z_curr = discount_factor_safe(s_curr, ...)
        Q_prev = survival_probability_pieewise(s_prev, ...)
        Q_curr = survival_probability_pieewise(s_curr, ...)
        prot += (Z_prev + Z_curr) * (Q_prev - Q_curr)
    return prot * (1 - ns_recovery) / 2
```

The RPV01 for the seasoned CDS includes four terms to handle the stub period between

t_{-1} and t_0 :

$$\begin{aligned}
\text{RPV01}(0, T) = & \underbrace{\Delta(t_{-1}, 0) Z(0, t_0) (1 - Q(0, t_0))}_{\text{Term A: accrued premium from } t_{-1} \text{ to today, upon default before } t_0} \\
& + \underbrace{\frac{1}{2} \Delta(0, t_0) Z(0, t_0) (1 - Q(0, t_0))}_{\text{Term B: accrued premium from today to default time, approximated}} \\
& + \underbrace{\Delta(t_{-1}, t_0) Z(0, t_0) Q(0, t_0)}_{\text{Term C: full coupon at } t_0, \text{ conditional on survival}} \\
& + \underbrace{\frac{1}{2} \sum_{n=1}^N \Delta(t_{n-1}, t_n) Z(0, t_n) (Q(0, t_{n-1}) + Q(0, t_n))}_{\text{Term D: remaining quarterly coupons}} \tag{8}
\end{aligned}$$

This is implemented in `rpv01_seasoned()`. The clean MTM deducts accrued interest:

$$\text{Clean MTM} = V(0) - S_{-t} \times \Delta(t_{-1}, 0) \tag{9}$$

The full computation is in `mtm_seasoned_cds()`.

2 Results

2.1 Interest Rate and Discount Factor

Table 2 shows the Nelson-Siegel interest rate and discount factor at integer maturities $t = 1, 2, \dots, 30$ years. The interest rate rises from near zero at $t = 1$ to approximately 3.59% at $t = 30$, while the discount factor declines from 0.999 to 0.340.

Table 2: Interest rate $\hat{r}(t)$ and discount factor $Z(0, t)$

t	$\hat{r}(t)$	$Z(0, t)$	t	$\hat{r}(t)$	$Z(0, t)$
1	0.000694	0.999307	16	0.031654	0.602626
2	0.003604	0.992817	17	0.032190	0.578546
3	0.007574	0.977534	18	0.032668	0.555423
4	0.011559	0.954817	19	0.033096	0.533221
5	0.015151	0.927045	20	0.033481	0.511906
6	0.018236	0.896358	21	0.033829	0.491441
7	0.020829	0.864330	22	0.034146	0.471795
8	0.022990	0.832005	23	0.034435	0.452933
9	0.024790	0.800030	24	0.034700	0.434825
10	0.026295	0.768780	25	0.034944	0.417442
11	0.027562	0.738461	26	0.035170	0.400753
12	0.028638	0.709171	27	0.035378	0.384731
13	0.029559	0.680949	28	0.035572	0.369350
14	0.030354	0.653798	29	0.035752	0.354584
15	0.031046	0.627699	30	0.035920	0.340408

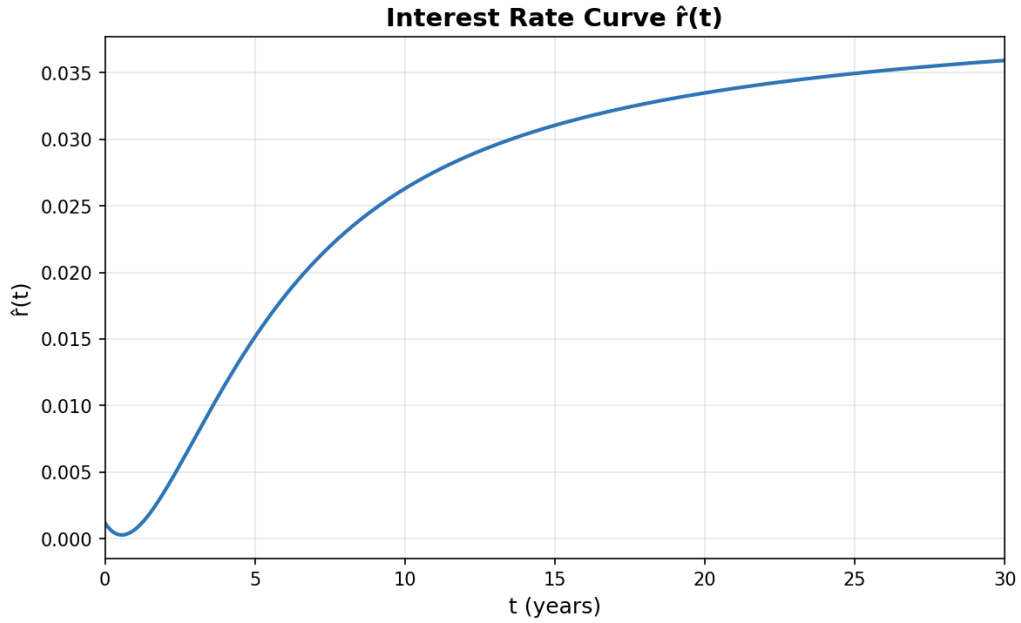


Figure 1: Interest rate curve $\hat{r}(t)$

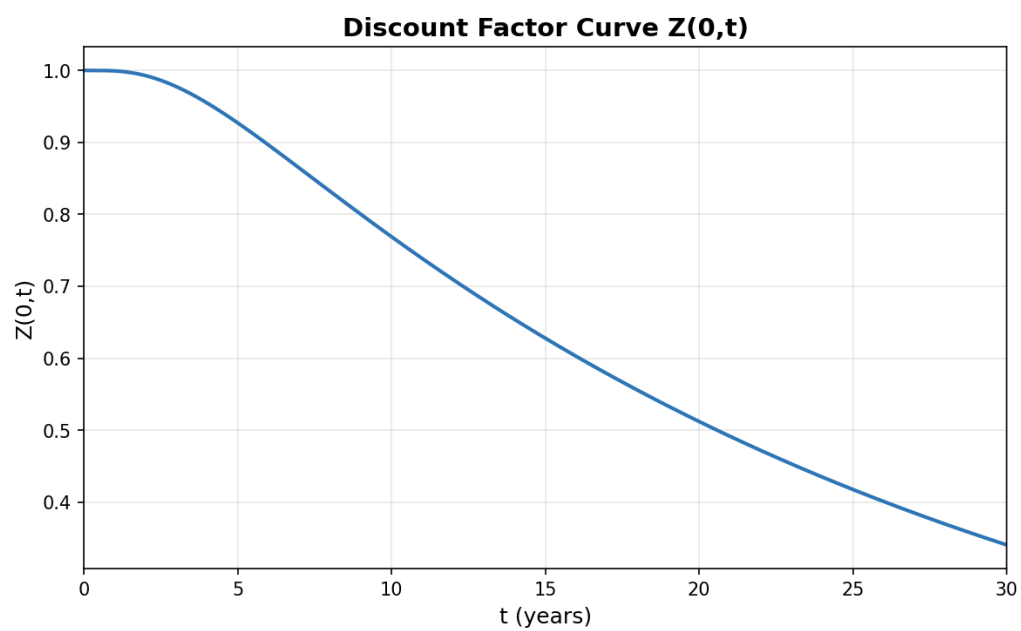


Figure 2: Discount factor curve $Z(0,t)$

2.2 Bootstrapped Forward Default Rates

Table 3 presents the bootstrapped forward default rates h_i . All residuals $V_i(h_i)$ are effectively zero (order 10^{-7} to 10^{-12}), confirming accurate calibration to market CDS spreads. Newton-Raphson converges in 3–4 iterations for each h_i .

Table 3: Bootstrapped forward default rates h_i

Tenor	Maturity (yrs)	Spread (bp)	h_i	$V_i(h_i)$	Iterations
6M	0.5722	103.07	0.018740	-4.24×10^{-11}	3
1Y	1.0694	104.68	0.019369	-3.16×10^{-11}	3
2Y	2.0889	106.74	0.019791	-6.47×10^{-10}	3
3Y	3.1000	110.31	0.021416	-1.11×10^{-9}	3
4Y	4.1139	113.38	0.022386	-1.48×10^{-9}	3
5Y	5.1278	116.98	0.024139	-2.38×10^{-9}	3
7Y	7.1639	128.78	0.029763	-1.27×10^{-7}	3
10Y	10.2028	143.51	0.034542	-1.89×10^{-12}	4
20Y	20.3472	159.43	0.034676	-8.20×10^{-9}	4
30Y	30.4972	164.13	0.035232	-4.50×10^{-9}	4

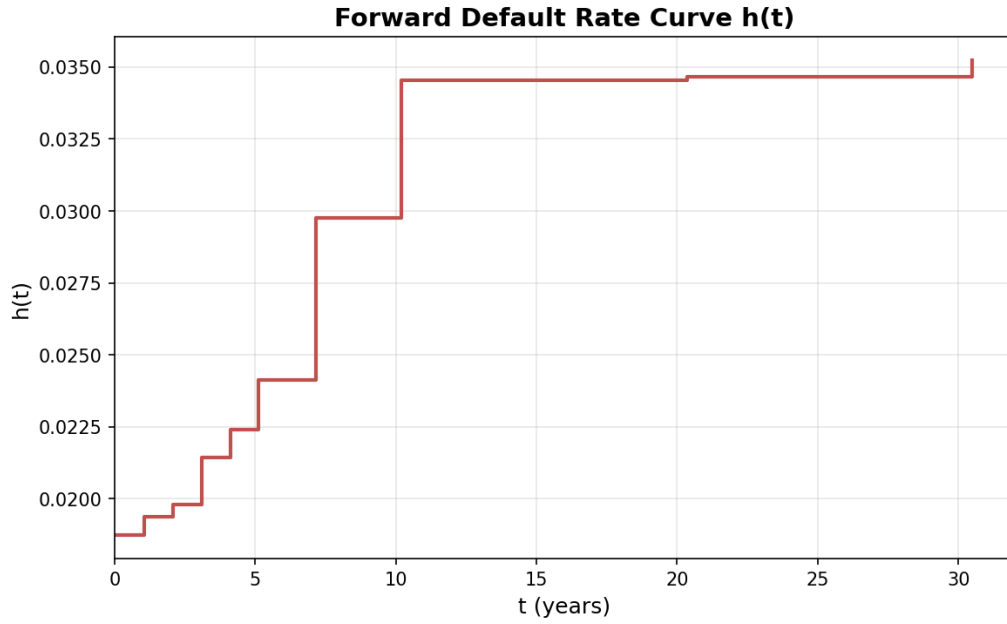


Figure 3: Forward default rate curve $h(t)$

2.3 Calibrated Survival Probability Curve

Table 4 shows the calibrated survival probability $Q(0, t)$ at integer maturities. The survival probability declines from 1.0 at $t = 0$ to approximately 0.380 at $t = 30$.

Table 4: Calibrated survival probability $Q(0, t)$

t	$Q(0, t)$	t	$Q(0, t)$	t	$Q(0, t)$
0	1.000000	11	0.738918	22	0.504130
1	0.981171	12	0.713735	23	0.486678
2	0.961971	13	0.689410	24	0.469830
3	0.941724	14	0.665914	25	0.453565
4	0.920967	15	0.643218	26	0.437863
5	0.899181	16	0.621297	27	0.422705
6	0.873441	17	0.600122	28	0.408072
7	0.847828	18	0.579669	29	0.393945
8	0.819684	19	0.559913	30	0.380307
9	0.791854	20	0.540830		
10	0.764969	21	0.522209		

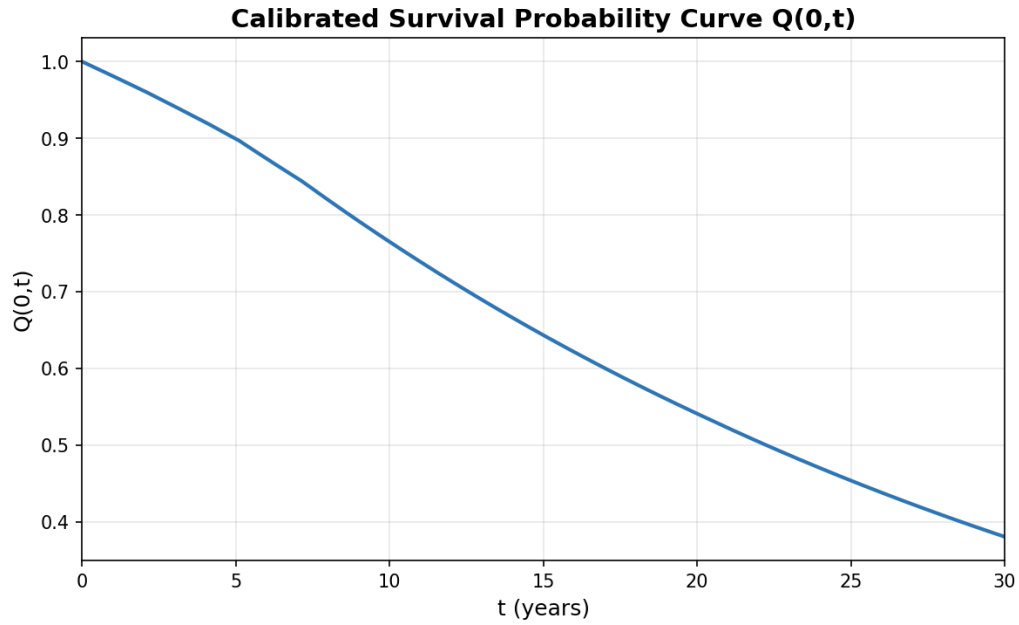


Figure 4: Calibrated survival probability curve $Q(0, t)$

2.4 Mark-to-Market Value of Non-Standard CDS

The non-standard CDS has the following key dates: the last payment date before valuation is $t_{-1} = 17$ February 2014 (rolled from 15 Feb, a Saturday), and the first future payment date is $t_0 = 15$ May 2014. There are 113 future payment dates in total. Table 5 summarises the pricing results.

Table 5: Mark-to-market results for non-standard CDS (face value USD 100 million)

Item	Value (USD)
Protection Leg	\$16,599,748.22
Premium Leg ($S_{-t} \times \text{RPV01}$)	\$24,138,762.13
Full MTM (Protection – Premium)	–\$7,539,013.91
Accrued Interest	\$51,944.44
Clean MTM (Full MTM – Accrued)	–\$7,590,958.36

3 Discussion

3.1 Interest Rate and Discount Factor Curves

The Nelson-Siegel interest rate curve is upward-sloping, starting near zero at short maturities and approaching approximately 3.59% at 30 years. This shape is typical of a normal yield curve environment and reflects the term premium demanded by investors for longer-duration exposure. The very low short-term rate (0.07% at $t = 1$) is consistent with the near-zero interest rate environment prevailing in early 2014.

The discount factor curve declines smoothly from near 1.0 at short maturities to about 0.34 at 30 years. The steeper decline at intermediate maturities (roughly years 3–10) reflects the region where interest rates are rising most rapidly.

3.2 Forward Default Rates and Survival Curve

The bootstrapped forward default rates increase monotonically from approximately 1.87% per year for the shortest maturity to 3.52% for the 30-year segment. This upward-sloping hazard rate curve implies that the market prices in increasing instantaneous default risk at longer horizons. The steepest increase occurs between the 5Y (2.41%) and 7Y (2.98%) segments, suggesting a structural change in perceived credit risk at that horizon.

The calibrated survival probability at 30 years is approximately 38%, meaning the market implies a roughly 62% cumulative probability of default over 30 years. The Newton-Raphson algorithm converges rapidly (3–4 iterations) for all maturities, and all residuals are negligibly small (order 10^{-7} or better), confirming an accurate fit to market spreads.

3.3 Shape of the Forward Default Rate Curve

Several factors may explain the upward-sloping shape of the forward default rate curve. First, longer time horizons inherently carry more uncertainty about the issuer’s financial health, as the cumulative probability of an adverse event increases with time. Second, the slope may reflect a term premium: investors demand additional compensation for bearing default risk over longer periods, analogous to the term premium observed in interest rate curves. Third, for investment-grade issuers, short-term default is considered unlikely, but long-term prospects are harder to assess, naturally producing lower short-dated hazard rates relative to long-dated ones.

The steeper increase between the 5Y (2.41%) and 7Y (2.98%) segments is notable. This could indicate a market expectation of increased risk at that horizon—for example, upcoming

debt maturities, regulatory changes, or sector-specific concerns that become more likely over a medium-term window. Beyond 10 years, the forward default rate flattens at around 3.5%, suggesting that the market views very long-term default risk as roughly constant once the medium-term uncertainties are accounted for.

3.4 Mark-to-Market of Non-Standard CDS

The full MTM of $-\$7,539,014$ indicates that the protection buyer is currently losing on this contract. This is because the contractual spread of 170 bp, agreed at inception in May 2012, significantly exceeds the current fair spread implied by the calibrated survival curve. In other words, the protection buyer is overpaying relative to current market conditions, suggesting that the reference entity's perceived credit risk has improved since the contract was initiated.

The high recovery rate assumption of 60% for the non-standard CDS (versus 45% for the standard CDS used in calibration) also contributes to the negative MTM. A higher recovery rate reduces the net payout upon default to only 40% of face value, directly reducing the value of the protection leg.

The accrued interest of $\$51,944$ reflects only 11 days of premium accrual since the last payment date (17 February 2014). This is small relative to the overall MTM but is necessary to compute the clean MTM. The clean MTM ($-\$7,590,958$) is slightly more negative than the full MTM because the accrued premium (owed by the buyer) is deducted.

3.5 Sensitivity to Recovery Rate Assumption

The recovery rate assumption materially affects the valuation. The non-standard CDS uses $R = 60\%$, compared to $R = 45\%$ for the standard CDS used in calibration. A higher recovery rate reduces the protection leg payout upon default from 55% to 40% of face value, directly scaling down the value of the protection.

If we were to price the non-standard CDS with $R = 45\%$ instead of 60%, the protection leg would increase by a factor of approximately $(1 - 0.45)/(1 - 0.60) = 1.375$, or 37.5%. This would make the MTM significantly less negative for the protection buyer, since the protection leg would be closer in magnitude to the premium leg.

This highlights that the recovery rate is not merely a technical input—it is one of the most influential parameters in CDS valuation. In practice, the appropriate recovery rate depends on the seniority of the reference obligation, the jurisdiction, and the industry sector. The choice of $R = 60\%$ for this contract (versus 45% for the standard market convention) suggests that the reference obligation may be senior secured or otherwise expected to have above-average recovery in the event of default.

3.6 Internal Consistency

The model is internally consistent: the survival curve is calibrated to reproduce all 10 market CDS spreads to high precision, and the non-standard CDS is then priced using the same calibrated curve and discount factors. The negative MTM is economically intuitive given that the contractual spread (170 bp) substantially exceeds the market fair spreads for comparable maturities.